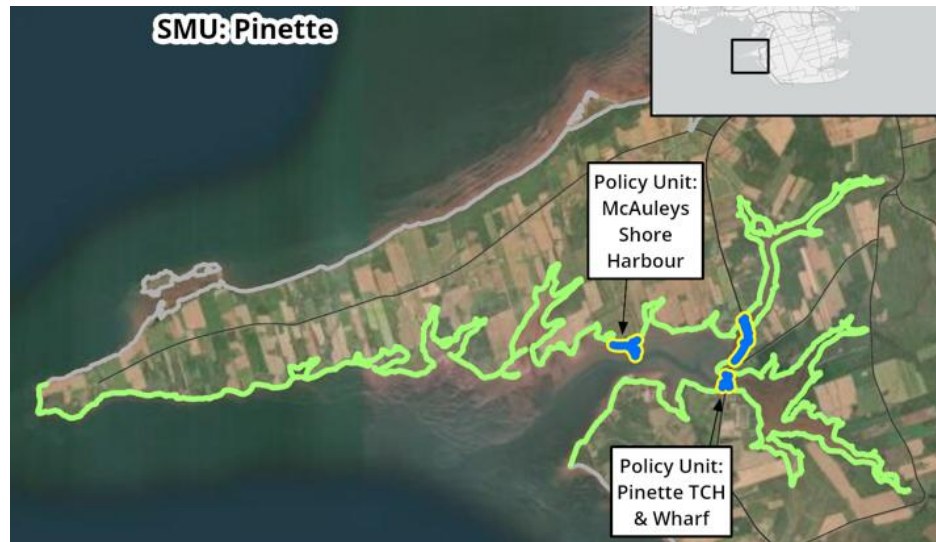


18. Pinette

Shoreline length: 67 km



Shore Types

Pinette is a tidal estuary with a south facing peninsula located on the southeast end of PEI, with a majority of low plains and wetlands.

% Shore types within SMU

low plain	bluff	sloped armour	cliff	wetland	vertical struct.	dunes	sandspit
51	2	6	2	38	0	0	1

Governance, Infrastructure, and Economy

This SMU is almost completely located within the Rural Municipality of Belfast (94%).

Key infrastructure includes a portion of the Trans-Canada Highway and provincial roads: Camp Buchanan Road (Mount Buchanan); McCauleys

Wharf; the TCH at Mount Buchanan Creek to Pinette Provincial Park; the TCH from Silver Waters Drive to the South Branch Pinette River; Gillis Road (Eldon); and the Pinette Wharf (South Branch Pinette River). Coastal economic drivers include the Pinette Raceway and McCauleys Wharf. Local and legacy land use constraints include several small lots and unpaved roads.



Point Prim Lighthouse



Bluff along Pinette Point south shoreline



Pinette wharf



TCH bridge

Natural Assets and Heritage

The Point Prim lighthouse is located on the western end of the area. The Pinette Provincial Park which consists of a playground and beach inside the estuary.

The shoreline includes critical habitat for bird species including the threatened Bank Swallow.

Planning and Policy Framework

Provincial rules apply. Land use and development is managed under provincial development standards, which include horizontal coastal building setbacks based on edge of feature and a calculation of erosion over time, building setbacks for dunes, and a requirement for a coastal buffer for new subdivisions. There are no specific flood elevations identified in regulations, but coastal hazard assessments are typically conducted.

Sediment transport

The estimated net sediment transport rate averaged across the SMU is moderate. Sediment transport rate potential is highest along the south facing peninsula.

Coastal Flood and Erosion Risk

The area has higher infrastructure density near Pinette Road and the Trans-Canada Highway, the remaining areas have limited development. Infrastructure in this area are vulnerable to increasing flood hazards. Erosion hazard is limited due to the sheltered orientation of the area.

The following flood risk indicator is based on the even with 1% annual exceedance probability (100-year return) with increasing amounts of sea level rise (SLR). The erosion risk indicator is based on potential future shoreline change for areas with historical data, not including salt marshes.

Estimated number of buildings and length of road at risk [per km shoreline].

	FLOOD RISK				EROSION RISK		
	Present	Short-term 0.3m SLR (2050)	Medium-term 0.6m SLR (2080)	Long-term 1.2m SLR (2130)	Short-term (~2050)	Medium-term (~2080)	Long-term (~2130)
Buildings	10 [<1/km]	17 [<1/km]	24 [<1/km]	46 [1/km]	5 [<1/km]	14 [<1/km]	29 [<1/km]
Roads	1298 m [20 m/km]	1551 m [23 m/km]	2129 m [32 m/km]	3521 m [53 m/km]	30 m [0 m/km]	162 m [2 m/km]	688 m [10 m/km]

Proposed SMP Strategies

	Short-term	Medium-term	Long-term
Sea level rise range	0 – 0.3 m	0.3 – 0.6 m	0.6 – 1.2 m
Potential timeframe	2030 - 2050	2050 - 2080	2080 - 2130
18.00 Pinette	NAI	MR	MR
Rationale	No Active Intervention is proposed for the short-term, followed by Managed Realignment. This will reduce risk to the limited number of vulnerable assets while preserving natural shorelines.		
18.01 McAuleys Shore Harbour -PU	R	R	R
Rationale	A Resist strategy is proposed for this key infrastructure. Future works would also require flood mitigation, subject to final strategies to be developed with the local authority.		
18.02 Pinette TCH and wharf -PU	R	R	R
Rationale	A Resist strategy is proposed for this key infrastructure. Future works would also require flood mitigation, subject to final strategies to be developed with the local authority.		

The table above shows how the recommended shoreline management unit (SMU) or policy unit (PU) strategy evolves across three planning epochs (Short-, Medium-, and Long-term) under increasing sea-level rise and erosion projections.

Adaptation strategies are preliminary recommendations subject to further development with feedback from community and government. These strategies are not approvals, commitments or policy determinations.